

ORIGINAL ARTICLE

Should I stay or should I go? Embracing causal heterogeneity in the study of pandemic policy and citizen behavior

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Abstract

Objective: To test for multicausality between government policy, health outcomes, economic performance, and citizen behavior during the COVID-19 global pandemic.

Methods: We perform Granger-causality tests to explore the interrelationship between four endogenous variables, social distancing policy, home isolation, balance rate, and average weekly COVID-19 deaths, in the 26 states of Brazil. As exogenous variables, we included a linear time trend and a dummy for the week in which the World Health Organization (WHO) declared COVID-19 a global pandemic.

Results: Our analysis of Granger causal ordering between the four variables demonstrates that there is significant heterogeneity across the Brazilian federation. These findings can be interpreted as underscoring that there is no common model applicable to all states, and that the dynamics are context-dependent.

Conclusion: Our suggested approach allows researchers to account for the complex interrelationship between government policy, citizen behavior, the economy, and COVID-related health outcomes.

KEYWORDS

COVID-19, Granger causality, government policy, political economy

The COVID-19 pandemic has affected the daily life of almost every person in almost every nation. In the face of a major public health crisis and the ensuing economic hardship, every government has had to make difficult decisions about policies and regulations regarding how to suppress COVID-19 and mitigate its impact (Lipsy, 2020). Citizens, in turn, must adapt to ever-changing information about the risks posed by the virus in their local communities and make difficult decisions about whether to comply with these regulations (Shaw, Kim, and Hua, 2020). Governments have to weigh whether curtailing economic activity must be prioritized to save lives (Apedo-Amah et al., 2020) and the tradeoffs in the number of infections that can be averted from school closures against the longer term costs for children. In countries with competing sets of government actors and policy environments, decisions about suppression and compliance

have been made considerably more complicated. Lockdowns are quickly enacted, but often backtracked in response to business pressures and citizen revolts, only to be quickly reintroduced as cases and deaths rise further (Adolph et al., 2021). Subnational governments that appear to be facing similar rates of COVID-19 deaths and cases often choose to enact very distinct policy responses.

To date, most of the articles examining these issues during the COVID-19 pandemic have assumed a single causal path from government policies to citizens to health outcomes. This approach may be appropriate, especially in the short-term to understand the dynamics at the onset of the pandemic and the effect specific interventions can have on reducing transmission while health infrastructure is prepared (Hsiang et al., 2020). However, strong arguments can be made that reverse the causal arrows implicit in the extant approaches, especially as what was understood as an initial and urgent emergency has evolved into a prolonged crisis. In scenarios where the electoral consequences of political actions are difficult to anticipate, politicians might have an eye on citizens' willingness to follow government policies as much as citizens look to the government for guidance. Moreover, citizens and leaders will adjust their responses over time as they learn from the immediate past, and form expectations about likely trajectories in the medium term. We thus think that it is imperative to use research designs in which we relax strong assumptions about exogeneity and allow for the possibility that causality flows both ways or even in the reverse direction of what is being assumed in much of the early research on the pandemic.

It is also important to develop models of the pandemic that take into account the political and economic circumstances in which governments and citizens are making their decisions about how to respond to a massive public health crisis when many of the key parameters are uncertain.¹ In countries where the governing structure is highly centralized, it may make sense to model trends at the national level. But, given the decentralized nature of many government structures and the nature of the pandemic, we make a case for theorizing about and empirically modeling the pandemic at the level at which policies are being set (Dodds et al., 2020). We also make a case for taking closer account of the economic environment and its role in shaping government policies and citizen behavior.

In this article, we explore the ways in which the severity of the pandemic, government policy responses, economic factors, and citizen behavior all shape each other. Given that reasonable theoretical arguments can be made for reciprocal causal relationships between any pair of these four factors, we make a case for the use of research designs and empirical testing in which addressing questions of endogeneity is the first step.² A corollary of our approach is to show that the modeling of these endogenous interactions must be considered within the context of how these patterns are evolving across time, and the challenges posed by inferring patterns when phenomena are characterized by their sudden explosiveness and highly volatile nature, as often occurs during pandemics.

Our results show that our understanding of COVID-19 dynamics is greatly enriched by examining the complex interactions among the multiple explanatory factors at work and understanding that these interactions vary depending on which state actors are leading the pandemic response. In one example from our analysis, the stark increases in COVID-19 deaths yields future changes in the stringency of social distancing policies, but the reverse is also true. Social distancing policies affect future death rates. In other examples, however, we show that the effect of death rates does not depend on government policies, and, vice versa. In other words, even if policies are changing intermittently within each state, they may not have the same impact in each locality.

In the sections that follow, we begin with a discussion of the relevant literature. We then discuss competing theories about how public health outcomes, government policies, citizen behavior, and economic factors influence each other. We follow this with a discussion of the research design and data that we use to

¹ For example, there has been considerable dispute about how COVID-19 is transmitted, the symptoms it causes, the period it takes for symptoms to emerge and subside. As a result, the World Health Organization's (WHO's) documents on the pandemic have undergone numerous revisions. For the latest version, see www.who.int/emergencies/diseases/novel-coronavirus-2019/question-and-answers-hub/q-a-detail/coronavirus-disease-covid-19.

² In this article, we employ the term "endogeneity" to mean multidirectional causality or reversed causality. We are aware that this term is most often employed by econometricians to refer to violations of Gauss–Markov conditions in the context of OLS. Our reference to endogeneity should not be interpreted in this fashion.



FIGURE 1 Classical approach

empirically test our competing theoretical claims on data measured at the state level in Brazil. We present our findings and then provide a discussion of our conclusions and some possible directions for future research on these important topics.

MODELING THE CAUSES AND EFFECTS OF POLICY DURING THE COVID-19 PANDEMIC

To date, the dominant theoretical approach to study the COVID-19 pandemic has been based on standard mathematical simulations commonly employed to describe and analyze epidemiological processes. The well-established susceptible-exposed infectious-recovered (SEIR) model is canonical. This modeling strategy assumes a unilateral causal order summarized by Figure 1 in which government policies shape citizen behavior, which then shapes health outcomes. As a result, the majority of studies that have analyzed the contribution of policies to reducing the spread of SARS-CoV-2, which include closing educational institutions, limiting gatherings, stay-at-home orders, lockdowns, and closing national and international borders (e.g., Brauner et al., 2021; Dehning et al., 2020; Maier and Brockmann, 2020; Kraemer et al., 2020; Chinazzi et al., 2020; Pei, Kandula, and Shaman, 2020; Armstrong, Lebo, and Lucas, 2020; Hsiang et al., 2020; Flaxman et al., 2020; Lai et al., 2020; Krishnamachari et al., 2021) have assumed that these interventions reduce individuals' mobility and hence the number of new cases and deaths.

As the pandemic spread across the globe in 2020, however, it quickly became apparent that there were no policies or solutions that would quickly eradicate the disease (Kissler et al., 2020). The examples of countries such as China and New Zealand that were able to reach a low level of cases after imposing strict lockdowns were rare. Instead, in countries such as in the United States and Brazil, governments lifted social distancing policies quickly or never activated them nationwide (Adolph et al., 2021; Barberia et al., 2021). Intermittent ("on," "off," "on" again) social distancing policies, were increasingly recognized to be politically easier to implement and enforce, though also recognized to impose larger overall economic or health-care costs as these policies only shifted the infection peak into the future (Rachel, 2020). Scholars also began to analyze how heterogeneity between regions within countries was essential to understanding the spread of the epidemic and to design effective strategies to control the disease (Della Rossa et al., 2020).

However, one of the shortcomings of these studies is that they often fail to consider that citizen behavior and disease spread may influence policy (Papageorge, 2021). In this article, that is the direction we take. Given the heterogeneity in the dynamics of the pandemic across states in the Brazilian federation, we explore the possibly mutual relationship between government policy, citizens' compliance to such policies, health outcomes, and economic constraints. Our results thus shed light on the intricate and context-dependent relationship between governments, citizens, the economy, and COVID-related deaths.

THEORY

In some ways, the COVID-19 pandemic is unlike any other challenge that governments throughout the world have faced in modern times.³ The scale of the dual crises in terms of threats to public health and

³ Stasavage (2020) argues that the current pandemic has many parallels with past threats posed by war, famine, and pestilence. While we agree that there are many similarities with past crises, the speed of transmission of information and the virus in the current pandemic is distinct. For our purposes, this is important as the endogeneity of the current pandemic occurs at an even more intense rate within and across countries.

economic well-being is certainly monumental. On the other hand, though, there have been large-scale economic and health crises in the past and there has been substantial research on the ways in which governments respond to them.⁴

Much of the work on government policy responses to crises and the work on government policies more generally begins with the assumption that democratic governments always have one eye on public opinion and the next elections as they craft their policies (Stasavage, 2020; Wlezien, 1995; Soroka and Wlezien, 2010; McGann, Dellepiane-Avellaneda, and Bartle, 2019). Given the abundance of evidence that public opinion shapes policies as much as policies shape public opinion, it certainly makes sense to explore the possibility of a mutual relationship between citizens' opinions and behavior, and policies during the COVID-19 pandemic. Thus far, there has been extensive research showing that citizens responded to government policies in the COVID-19 pandemic. However, information on the reverse causal arrow is lacking. By investigating whether policymakers react to citizens' behavior, we deepen our knowledge about the variation in the adoption of specific policy measures across governments during COVID-19.

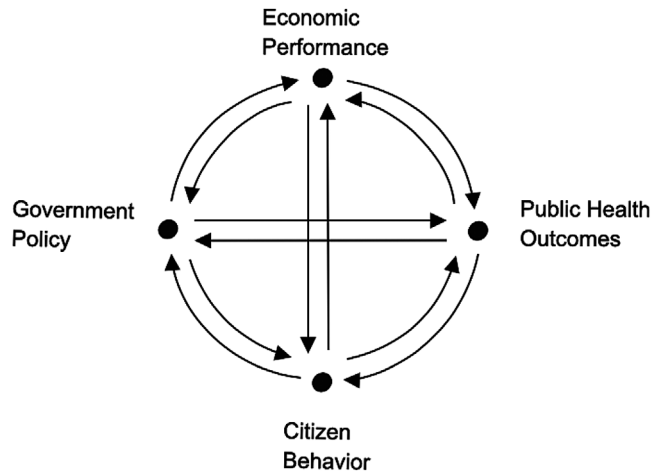
This is especially relevant given the lengthy duration of the pandemic. In these types of health crises, similar to most other prolonged emergencies, public policies are often made with significant uncertainty as to their effects (Manski, 2013). The high level of uncertainty and extreme time pressure means that governments and society are continually learning and adapting to a new environment in a series of strategic interactions. Policies are not one-time interventions; they are being shaped in an iterative fashion based on scientific research that is being developed in real time, and subjective perceptions of risk. In these successive scenarios where the consequences of a set of policies are particularly difficult to anticipate, governments should be even more prone to search for cues on how citizens and economic actors are responding to the enacted measures and to adjust accordingly. In turn, whereas the public may have had an initial "rally around the flag" reaction at the onset of the COVID-19 pandemic (Yam et al., 2020), attitudes have shifted in the successive waves that have occurred in most countries (Arnot et al., 2020).

Needless to say, governments' decisions on which policies to put in place to contain the pandemic and citizens' decision to reduce mobility and physical contact so as to reduce the COVID-19 transmission rate depend on more than just each other, and also involve assessments of the costs and benefits of policy choices at different points in time (Soofi, Najafi, and Karami-Matin, 2020). The severity of the pandemic in terms of health outcomes (Schraff, 2020) and the short- and long-term economic consequences (Jordà, Singh, and Taylor, 2020) associated with stringent policies all play into citizens' and governments' decisions regarding the costs and benefits of strict policy measures. In a struggling economy, citizens' concerns over an immediate economic loss and even of a loss of a means of livelihood might outweigh the perceived risk associated with infection by COVID-19 (Soofi, Najafi, and Karami-Matin, 2020). On the other hand, as the risk of infection and death increases, citizens might feel more inclined to support social distancing policies. Governments, in their turn, must anticipate how social distancing policies will affect the economy, and whether that will lead to a backlash from citizens. Moreover, they must also be attentive to the negative repercussions of an increase in the number of infections and deaths, and the possibility of overwhelming the health system.

Heterogeneity also matters. In many countries, there are compelling reasons to examine trends at the subnational level. Capano (2020), for example, notes that national-level coordination of agencies in terms of crisis management (disasters, earthquakes, etc.) were not deployed in the case of Italy during the COVID-19 pandemic. Instead, regions emerged as the central actors. Italy is one such example where subnational governments were largely responsible for setting policies within their jurisdictions (Migone, 2020). However, whereas some federations, such as Canada, adopted a cooperative response with the national government to the pandemic, there are other cases, such as the United States and Brazil, where

⁴ On economic crises, see, for example, Gourevitch (1986) and Bermeo and Pontusson (2012). On the AIDS pandemic as an example of a health crises, see Boone and Batsell (2001), Baldwin (2005), De Waal (2003), Dionne (2011), Lieberman (2009) and on the more recent 2002–2004 SARS-CoV-1 outbreak, see Lai (2018).

FIGURE 2 Possible relationships in our theoretical approach



the interplay between these levels of government has been contentious. In these cases, states that appear to be under the same pandemic scenario often respond quite differently at similar pandemic peaks and ebbs.

In addition to subnational regions often being the appropriate level for analyses of the effects of pandemic policies, there are advantages to conducting comparative studies across subnational regions. In particular, such studies hold constant a large number of variables that do not vary or do not vary much within nations. These include variables measuring factors such as national politics, culture, and, in many cases language and history. Of course it is also important in a study of subnational regions over time to include variables that measure relevant national-level factors, such as changes to federal government policies, which vary across time within the period studied.

To capture the dynamics of the current pandemic, we propose an approach that allows for the interrelationship between citizens' behavior, governments' actions, the economy, and health outcomes. Importantly, we emphasize that any meaningful account of the causes and consequences of government policies must take into account the level at which policy is predominantly being made. To be specific, assuming that national policies are reflective of the policy environment at lower levels of government is inadequate. In many countries, major decisions on how to contain the spread of COVID-19, including closing nonessential services and mandatory mask orders, were made and enforced by local governments. In fact, the current pandemic has revealed that the multitude of actors associated with a decentralized power structure can often produce contradictory guidelines between central and local governments (Capano, 2020).

In Figure 2, we summarize our theoretical ideas about how government policy, citizen behavior, economic performance, and public health outcomes interact and influence each other's actions during the pandemic. The arrows from a given variable to another variable indicates a causal temporal order. There are a total of 12 arrows that indicate that bidirectional relationships may be present between all four of the major factors in the current crisis.

RESEARCH DESIGN

To test the theoretical propositions that we discuss in the previous section, we need an empirical modeling strategy that allows for causal relationships to flow between multiple pairs of possibly endogenous variables, while allowing the identification of causal directionality in a context where nonstationarity cannot be resolutely rejected for some of the variables in the model. We also need time series measures of all four of the variables represented in Figure 2 at the level at which the political action is taking place.

Modeling strategy

A well-known and frequent approach to identifying causal precedence, which was first proposed by Granger (1969), is based on the premise that such an order exists if it can be shown that earlier values of a variable, X , help to predict future values of another variable, Y . In the case of a simple bivariate model, the so-called Granger-causality test employs a t -test to verify if the β coefficient on the lagged value of X is statistically different from zero. In the case of a VAR(1) (where VAR is vector autoregressive), as summarized by the system of equations:

$$\begin{aligned} x_t &= \beta_{11}x_{t-1} + \beta_{12}y_{t-1} + e_{1t} \\ y_t &= \beta_{21}x_{t-1} + \beta_{22}y_{t-1} + e_{2t}. \end{aligned} \quad (1)$$

There are two hypotheses that are tested ($\beta_{12} \neq 0$ and $\beta_{21} \neq 0$). In the case that there are more lags included, the test employs a joint F -test or Wald test to test the null that all the coefficients of the lagged variables are equal to zero. If the null hypothesis that $\beta_{12} \neq 0$ can be rejected, we can conclude that Y Granger causes X . If the null hypothesis cannot be rejected, we can conclude that there is no statistical evidence to suggest that Y Granger causes X .⁵

Toda and Yamamoto (1995) introduce a procedure for conducting Granger-causality tests (Granger, 1969), which avoids potential biases that result from inconclusive unit root (Campbell and Perron, 1991) and cointegration tests (Maddala and Kim, 1998) and allows us to utilize Granger-causality tests without having to differentiate the possibly nonstationary series. This is all carried out within the context of standard VAR modeling framework. To implement the strategy outlined by Toda and Yamamoto (1995), we first specify the endogenous and the exogenous variables, then we (a) conduct integration tests to determine the maximum order of integration among all variables in the model (m)⁶; (b) use standard information criteria to identify the appropriate lag length for the endogenous variables in the VAR (p); in this case, we adopted the Schwarz information criterion; (c) estimate VAR models in levels; (d) check for serial correlation in the residuals and if the system is dynamically stable; and (e) estimate the VAR model adding in the m lag of each of the endogenous variables as exogenous coefficients, VAR($p + m_{max}$);⁷ and, (f) evaluate the hypotheses for the Granger-causality tests with the test statistics correctly adjusted for asymptotics.

The work of Lütkepohl (1982) was instrumental in calling attention to the bias caused by conducting Granger-causality tests when variables are omitted. Our empirical approach is thus designed to explore the extent to which there is multicausality between each pairwise combination of government policy, health outcomes, economic performance, and citizen behavior while controlling for the other possibly endogenous factors. There are thus a total of 12 possible bivariate Granger causal relationships in each of our models. We include two exogenous variables in each of our VAR models.⁸ These are a linear time trend to account for the trending patterns observed in some of the endogenous variables, and a dummy variable identifying epidemiological week 11, which includes March 11, 2020, the date on which the WHO declared COVID-19 a global pandemic.

⁵ Although there is much heated philosophical debate about whether it is even possible to establish causal relationships (Sims, 1980; Granger, 1980), evidence of Granger noncausality is widely accepted as the best first step toward establishing the temporal relationship with time series data. However, as Brandt and Williams (2006) point out, a finding of Granger noncausality does not imply that the series are noncorrelated. As they emphasize, the two series may be highly contemporaneously correlated.

⁶ For instance, if y_t and x_t were $I(0)$ and z_t $I(1)$, $m = 1$ for y_t , x_t , and z_t .

⁷ The inclusion of these extra coefficients (m lags) is a key part of the Toda and Yamamoto (1995) strategy to produce appropriate diagnostics in the case of possibly nonstationary variables (the m lags are treated as exogenous variables in Granger-causality tests, therefore, they are not considered as endogenous variables in the test).

⁸ The p -values reported in our results are the result of a Wald test for the explanatory value of the past values of each variable in question on present values of the one endogenous variable holding constant the other two endogenous variables and the exogenous variables in each model. We are keenly aware that tests of Granger causality are closely linked with cointegration (Levendis, 2018). However, our contribution in this article is to present a statistically valid procedure to test if the Granger causal relationships between two-specific variables is bidirectional, unidirectional, or nonexistent.

Data

We test our theory using data for the states of the Brazilian federation. There are several reasons why an analysis of South America's largest country is well suited for the questions we seek to explore. To date, Brazil has been one of the hardest hit nations during the pandemic. From the outset of the diagnosis of the first infections in Brazil, state governments took a lead role in implementing social distancing policies to prevent the spread of COVID-19 within their territories, and these subnational governments have been consistently recognized as dominant in cases where more lax federal legislation introduced by President Jair Bolsonaro has attempted to circumvent the containment strategies adopted by state governors.⁹ As a result, each state has had autonomy to adopt its own containment strategy and there has therefore been considerable variation in terms of patterns of government policy, health outcomes, economic performance, and citizen behavior across the Brazilian federation.

For our empirical data, we collected measures of our four endogenous variables in each state of Brazil. For our measure of citizen behavior, we assess the willingness of individuals to stay at home during standard work and school hours based on the percentage of cell phones, which remained at the same geographical location during the day (from 6 a.m. to 10 p.m.) and overnight (10 p.m. to 6 a.m.). This measure, which we refer to as "Isolation" in the discussion of our findings, is made available by Inloco (Ferraz, 2020) and data are based on smartphone cell phone users who have installed commercial applications that allow location tracking in Brazil. These data are collected for municipalities, and in this study we use a measure that is aggregated at the state level.¹⁰

For our measure of government policy, we use a composite index of social distancing policies collected by the COVID-19 Government Response Tracker for the Brazilian Federation database (Barberia et al., 2020). This index is based on the coding of the stringency level and geographical scope of restrictions enacted by state governments with respect to school and work closures (with separate measures for the commerce and service sectors, and industry), measures that prohibit the use of public space for gatherings and those which prohibit gatherings in private spaces, and the decrees that strictly limit the mobility of the population, requiring, at many times, the individuals' permanence at home during determined periods. The six indicators are aggregated into a Social Distancing Policy Stringency Index that varies from 0 (no policy intervention) to 100 (maximum level of intervention).

For our measure of health outcomes, we use deaths due to COVID-19. Data on deaths due to COVID-19 has been a source of considerable difficulty globally as each country collects and reports the data differently. In the case of Brazil, the data on new deaths that we use in this study are based on recorded deaths on a specific date (vs. on data of occurrence as is recorded in other countries) as reported by state governments. The source, Brasil.io, publishes compiled indicators of cases and deaths based on the official epidemiological bulletins released by the 26 State Health Secretariats (Justen et al., 2020).

The economic measure employed in this article is an employment market measure known as the "balance rate." It represents the number of employed workers who lose their job each week net of those who find jobs in the formal labor market standardized by the size of the formal market workforce. We are well aware that this measure does not fully capture labor markets in Brazil given the significant share of informal workers, but we are not aware of comparable measures that are available with high frequency during the pandemic. These data were retrieved from the *Cadastro Geral de Empregados e Desempregados (CAGED)* of the Ministry of Economics. Despite not representing the entire labor market as it is restricted to the formal market, CAGED is historically considered as a good thermometer of labor market trends in general

⁹ In a series of rulings starting from April to June 2020, the Brazilian Supreme Court (*Supremo Tribunal Federal*) ruled that the federal government could not interfere in the policies adopted by state and municipal governments to combat the pandemic. For further information, see *Ação Direta de Inconstitucionalidade (ADI)* 6.341, ADI 6.343 and the *Arguição de descumprimento de Preceito Fundamental* (ADPF) 672.

¹⁰ Although the ownership of a smartphone in Brazil is indicative of being at least middle class, there is substantial evidence (Jay et al., 2020; Carlitz and Makhura, 2021) that citizens from all social classes follow similar trends, if not levels of isolation.

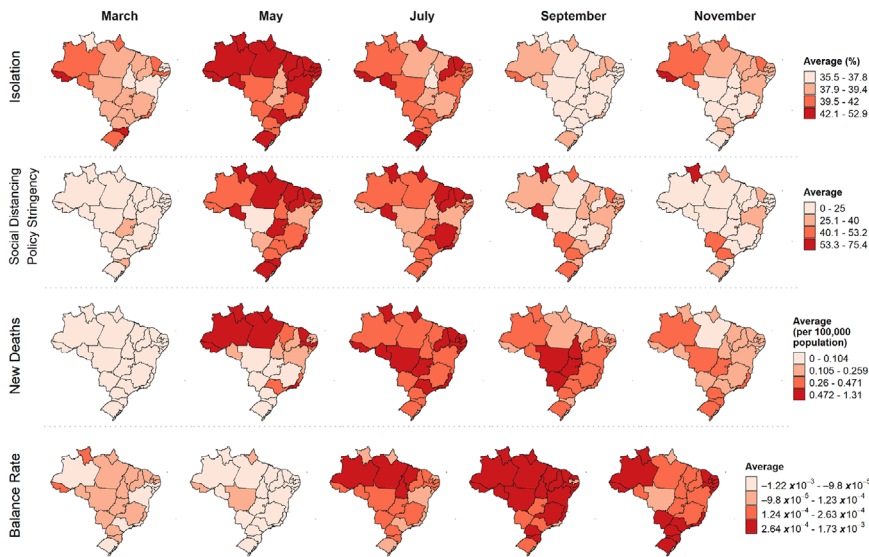


FIGURE 3 Monthly means of social distancing policy stringency, isolation, new deaths, and balance rate for the 26 states of the Brazilian federation, select months

in Brazil. The following formula explains how the weekly measure was calculated as a share of the stock of formal jobs (lagged):

$$\text{Balance rate} = \frac{\text{Job fires} - \text{Job hires}}{\text{Stock of formal labor jobs}}.$$

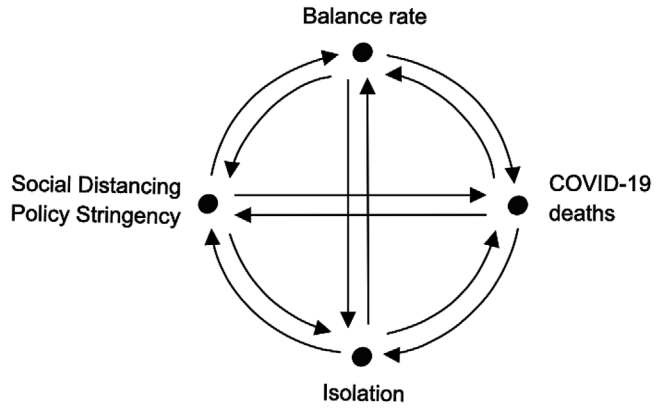
In order to diminish the problems with the substantial fluctuations that occur across days of the week, we estimate our models using weekly measures for the epidemiological weeks from 6 to 48 of 2020.¹¹ In addition, as is the case in most countries, social distancing policy changes are often enacted on a weekly or biweekly basis based on evaluations of 7-day and/or 14-day moving averages of the public health capacity and the epidemiological indicators in Brazil (Garcia et al., 2020).

The sample is restricted to the 26 states of Brazil. The Federal District of Brasilia is the only federative unit that is not formed by municipalities. This type of territorial differentiation produces significant administrative implications and it can affect how policies are designed. In order to compare only states with the same administrative and territorial divisions, we excluded the Federal District from the analysis. The summary statistics and the data sources are summarized in the first section of the Supporting Information.

Figure 3 presents the state averages for social distancing policy stringency, home permanence, new deaths, and labor market performance for select months in the sample. Although we employ weekly data in the study, we present monthly averages for the sake of brevity. The significant variation at the state level confirms why examining national outcomes can be potentially misleading in the case of Brazil. Although the month of May 2020 is the period with the highest levels of home permanence and social policy stringency, there are still a few states with high levels of citizen adherence to social distancing and social distancing policy stringency in later months. The pattern of new deaths confirms strong regional concentration of new deaths in the states of the North and São Paulo in May, and a more generalized spread after July, with reduction in some regions in specific points in time. Labor markets were contracting in the first months

¹¹ There are few COVID-19 deaths reported during the weekend. For this reason, on Mondays, the number of reported deaths are higher due to the weekend data backlog. In a similar fashion, individuals leave their homes at a higher proportion on weekdays, and even across weekdays, there are some days that have higher mobility.

FIGURE 4 Causality tests conducted for each state



of 2020, but there is a shift in the second semester with most states recovering jobs. However, similar to the other indicators, there is significant variation across states.

Summary of our research design

In Figure 4 we summarize the 12 causal relationships that we test in each state using our weekly data. For each pair of endogenous variables, A and B , we conduct two causality tests. The null hypothesis for each causality test is that there is no Granger causal relationship from endogenous variable A to endogenous variable B holding constant the two other endogenous variables and the two exogenous variables. There are thus four possible outcomes for the relationship between each pair of endogenous variables:

1. if we reject the null hypothesis for both tests, A to B and B to A , we have evidence of reciprocal causality denoted by a two-way arrow, $A \rightleftarrows B$, between these two variables in Figure 5);
2. if we reject the null hypothesis for the test of A to B , but not for the test of B to A , we have evidence of one way causality, denoted as $A \rightarrow B$;
3. if we reject the null hypothesis for the test of B to A , but not for the test of A to B , we have evidence of one way causality, denoted as $A \leftarrow B$;
4. if we are unable to reject either null hypothesis, we have no evidence of a causal relationship between variables B to A , denoted by the absence of any arrows between these two variables in Figure 5).

As our summary here emphasizes, our research design is oriented to robustly explore whether there are distinct multicausal patterns by examining the variation in the 26 states of the Brazilian federation. We view this approach as preferable to a pooled analysis. Nevertheless, it should be noted that we also considered using panel Granger-causality tests for verification. However, we were unable to obtain panel VAR stability, which is a necessary pre-condition for Granger-causality testing. In our view, however, panel Granger tests would be less useful as this test is designed to detect causal precedence at the panel level (Dumitrescu and Hurlin, 2012). Similar to other panel-specific tests, the rejection of the null hypothesis would not exclude noncausality for some states. Since we are interested in testing whether such heterogeneity exists across the states, this empirical approach would not be as helpful to our main objective.

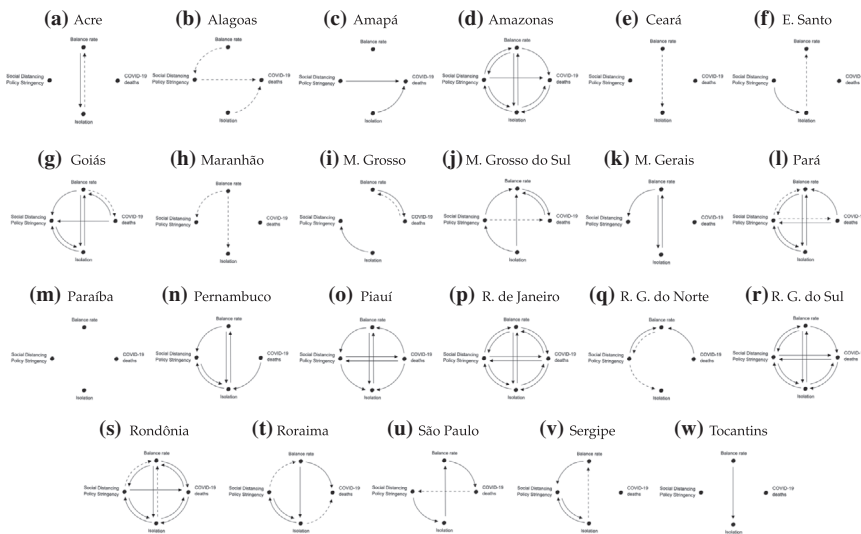


FIGURE 5 Granger-causality graphs. The endogenous variables included in the Granger-causality tests are represented by black nodes. While the solid lines indicate a 5 percent significance level, the dashed lines represent a 10 percent significant level. The null hypothesis of the Granger-causality test is that of no Granger causality between the two series. A rejection of the null suggests that there is Granger causality holding constant the additional endogenous variables. For the cases in which the Granger-causality test did not produce results that permit the null hypothesis to be rejected at 5 percent or 10 percent levels, no line is displayed. We also added as exogenous variables a time trend and a dummy variable for epidemiological week 11, which includes March 11, 2020, when the WHO declared COVID-19 a global pandemic

RESULTS

In Figure 5, we present graphical summaries of the results from the 12 Granger-causality tests that we conducted for each state.¹² The solid lines for arrows in Figure 5 indicate that the null hypothesis was rejected at the 5 percent or less significance level, while the dashed lines represent relationships for which the null hypothesis could only be rejected at a 10 percent significant level or less. These diagrams show substantial variation in the statistical significance and the direction of the Granger-causality tests for the 23 states.¹³ Indeed, it is difficult to find a common pattern across all states.

In the cases of Rio de Janeiro and Rondônia, for example, we found that all four endogenous variables Granger cause each other. However, in the case of Rondônia the effect of social distancing policy stringency on balance rate and the effect of isolation on balance rate are only statistically significant at the 10 percent level. Widespread endogeneity for all four variables is also detected in Amazonas and Rio Grande do Sul. As the figure shows, there is evidence of Granger causality in 10 and 11 of the total of 12 relationships in the cases of Amazonas and Rio Grande do Sul, respectively. In these states, the evidence suggests that there is bidirectional causality as a given variable Granger causes the other variable, and vice versa, in four (Amazonas) and five cases (Rio Grande do Sul). Only in the case of the balance rate to COVID-19 deaths is the relationship unidirectional for both states.

On the other end of the spectrum, another common pattern that we detect across multiple states is far less endogeneity among all 12 possible relationships. In the states of Ceará and Tocantins, a

¹² Before conducting the Granger-causality tests, we conducted a series of unit root tests to check the order of integration of each series. The results are reported in Tables A3, A4, A5, and A6. The detailed numerical results for the Granger tests for each state are reported in the Supporting Information Appendix.

¹³ For 3 of the 26 states, we were unable to run the Granger-causality tests. In those three cases (Bahia, Paraná, and Santa Catarina), we were either unable to purge serial correlation from the residuals or the VAR failed to satisfy the stability condition.

TABLE 1 Summary of results

Variable pair			$\rightleftarrows$	$\longrightarrow$	$\longleftarrow$	—	Total
Social distancing policy stringency	Isolation	<i>N</i>	9	4	2	8	23
		%	39.1	17.4	8.7	34.8	100
Social distancing policy stringency	COVID-19 deaths	<i>N</i>	4	5	2	12	23
		%	17.4	21.7	8.7	52.2	100
Social distancing policy stringency	Balance rate	<i>N</i>	6	2	7	8	23
		%	26.1	8.7	30.4	34.8	100
Isolation	COVID-19 deaths	<i>N</i>	4	3	2	14	23
		%	17.4	13.0	8.7	60.9	100
Isolation	Balance rate	<i>N</i>	10	4	4	5	23
		%	43.5	17.4	17.4	21.7	100
COVID-19 deaths	Balance rate	<i>N</i>	5	3	4	11	23
		%	21.7	13.0	17.4	47.8	100

Note: For each pair of endogenous variables, A and B , we conduct two causality tests. The null hypothesis for each causality test is that there is no Granger causal relationship from endogenous variable A to endogenous variable B holding constant the other endogenous variables and the two exogenous variables. In the table, we present the results when the null hypothesis was rejected at 10% level of less significance level. The symbol $\rightleftarrows$ (two-way arrow) denotes evidence of reciprocal causality between the two variables, $A \rightleftarrows B$. When we have evidence of one way causality for the test of A to B , but not for the test of B to A , we indicated as $A \longrightarrow B$. If we have evidence of one way causality for the test of B to A , but not for the test of A to B , we indicated as $A \longleftarrow B$. Finally, the absence of any arrows (—) indicates the case when we have no evidence of a causal relationship between variables A and B . N refers to the number of states and % indicates the percentage of the states that falls under the scope of one of the four categories ($\rightleftarrows$, $\longrightarrow$, $\longleftarrow$, or —).

common pattern is observed. For these states, we found evidence that there is only a single unidirectional case among the 12 possible relationships that were evaluated. In both states, the balance rate Granger causes isolation. However, this effect is only statistically significant at a 10 percent level for Ceará. In the other eleven relationships, the evidence did not suggest that either variable Granger causes the other.

In Table 1, we present a summary of the causality test results for each pair of endogenous variables across the 23 states for which we have results. From the outset, it should be noted that there is a rich variation across these cases. Starting at the top of this table, we can see that the results for the causes and consequences of government policies are heterogeneous. In contrast with standard approaches depicted in Figure 1, which assume a causal relationship between government policies and citizen behavior, we find the strongest evidence for a reciprocal relationship between these variables. This was found in nearly 40 percent of the Brazilian states. However, we also found no evidence of a causal relationship between these two factors in just over a third of states (34.8 percent). We found evidence supporting the assumed unidirectional causal relationship from policy stringency to isolation in only four states (17.4 percent) and evidence in favor of only the reverse causal arrow in two states (8.7 percent).

Moving to the second row of Table 1, we can see that our findings for causality tests from stringency policy to COVID-19 deaths are also mixed. In the majority of states (52.2 percent), we found that there was no causal relationship between these two variables. We found evidence of reciprocal causality between policy stringency and deaths in four states (17.4 percent), evidence of one-way causality from policy stringency to deaths in five states (21.7 percent), and evidence of one-way causality from deaths to policy stringency in only two states (8.7 percent). The results for the relationship between policy stringency and our measure of the economy (the balance rate) were also an interesting mix, with a fairly even spread of cases of reciprocal causality (26.1 percent), one-way causality from balance rate to policy stringency (30.4 percent), and no evidence of causality (34.8 percent). Somewhat surprisingly, we

only found evidence of one-way causality from policy stringency to the balance rate in only two states (8.7 percent).

Our findings for the second assumed causal relationship in Figure 1, from citizen behavior to health outcomes, are also quite mixed. As we can see from the four row in Table 1, the strongest pattern that we found from any of our causality tests was that isolation and deaths were not related to each other. We found this in 14 of 23 (60.9 percent) states. We found the implied one-way causal relationship from Figure 1 (from isolation to deaths) in only three states (13 percent), the reverse relationship in two states (8.7 percent), and reciprocal causality between isolation and deaths in four states (17.4 percent).

From the fifth row of Table 1, we can see that we found strong evidence of reciprocal causality between isolation and the balance rate (43.5 percent), followed by five states where there was no relationship (21.7 percent), and an even spread of states where there was a one-way causal relationship (17.4 in each direction). Finally, from the bottom row in Table 1, we can see that the dominant finding from our causality tests between the balance rate and deaths was a null finding (47.8 percent), followed by evidence of reciprocal causality (21.7 percent), one-way causality from the balance rate to deaths (17.4 percent), and one-way causality in the opposite direction (13 percent).

In light of these results, an obvious question is what explains the large variation in our causal findings across the Brazilian states. To try to answer this question, we attempted an inductive analysis using results from a multiple correspondence analysis of the findings summarized in Supporting Information Figure 5. We did a cluster analysis using hierarchical clustering on principal components (Husson, Lê, and Pagès, 2011; Husson et al., 2020). Full details of this analysis are provided in the Supporting Information of this study. The results show some suggestive evidence, but none of the conditions we test provide robust evidence that there is a single institutional or contextual factor that is driving the heterogeneity in causal precedence patterns.

CONCLUSION AND FUTURE DIRECTIONS

Our systematic and robust analysis of Granger causality between the economy, health outcomes, citizen behavior, and government policies reveals that there is significant heterogeneity across the Brazilian federation. There are some instances in which a state's environment exhibits a significant number of endogenous relationships that are bidirectional. In other cases, the degree of endogeneity is more limited and restricted to a subset of the four components. Together, these findings can be interpreted as underscoring that there is not a common model applicable to all states, and that the dynamics are context-dependent.

Our results are obviously limited to federal units with one nation and by the fact that we are covering only the first part of a crisis that may be with us for a substantially longer period of time. Despite these caveats, we think that the evidence that we have found sounds a strong warning against assuming rather than testing for causal relationships during the current pandemic. Given the iterative nature of all of the major factors, it makes sense to adopt research designs that allow for a heterogeneity of causal structures.

We have also made arguments for including measures of the economy in any model of the pandemic and for modeling the causes and consequences of government policies at the level at which policies are set. Initial lockdowns and other measures to slow the early spread of COVID-19 might have been made regardless of their economic consequences. But, as it has become clear that there would be no quick end to the pandemic, arguments about the economic consequences have become more salient. In some countries, decisions about restrictions have been consistently made at the highest levels of national government. In other settings, however, policies have been determined by lower level governments.

Moving forward, one thing that we know for certain is that we will have more data with which to study the causes and consequences of the current pandemic. This, in turn, will open up lots of opportunities for researchers to deepen our understanding of these incredible times. As an example of the possibilities this will generate, we will eventually be able to relax and test assumptions about causal heterogeneity across

time in addition to across spatial units as we have in the present study. We will also be better positioned to test for spatial diffusion relationships across units.

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SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section at the end of the article.

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